

Assignment Title: Investigation of Algorithm Efficiency in Real-Life Data Processing Systems

Course Outcome (CO)

CO3: Analyze the complexity of algorithms related to data structures to determine their efficiency and performance.

Program Outcome (PO)

PO4 – Investigation (C4: Analyze): Conduct investigations of complex problems using research-based knowledge and methods, including analysis and interpretation of data.

Learning Domain & Taxonomy Level

- **Cognitive Level:** C4 – Analyze
- **Aligned Topics:**
 - Time and Space Complexity
 - Sorting Algorithms: Bubble, Insertion, Selection, Quick, Merge
 - Searching Algorithms: Linear and Binary Search

Real-Life Application Scenario

Scenario: Online Food Delivery Order Management System

An online food delivery platform stores daily customer orders. Each order contains:

- Order ID
- Order amount
- Delivery distance
- Time of order

The system needs to:

- Sort orders for billing, analytics, and delivery optimization
- Search specific orders quickly during customer support or issue resolution

As a system analyst, you are required to **investigate which algorithms perform best under different real-world data conditions**.

Investigation Tasks

Task 1: Data Characterization

1. Assume three types of real-world data conditions:
 - Small dataset (≤ 100 orders)
 - Medium dataset ($\approx 1,000$ orders)
 - Large dataset ($\geq 10,000$ orders)
2. Identify whether the data is:
 - Random
 - Nearly sorted
 - Reverse sorted

Task 2: Sorting Algorithm Investigation

Investigate the performance of the following sorting algorithms:

- Bubble Sort
- Insertion Sort
- Selection Sort
- Quick Sort
- Merge Sort

For each algorithm:

1. Analyze **best, average, and worst-case time complexity**
2. Evaluate suitability for:
 - Small vs large datasets
 - Nearly sorted vs unsorted data
3. Identify practical limitations in real systems
4. Compare time and space complexity trade-offs

Task 3: Searching Algorithm Investigation

1. Analyze **Linear Search** and **Binary Search** for order lookup
2. Investigate:
 - Performance on sorted vs unsorted order lists
 - Time complexity differences in real usage scenarios
3. Justify when preprocessing (sorting) is worthwhile before searching

Task 4: Comparative Analysis (Critical Thinking)

- 1. Create a comparison table summarizing:
 - o Dataset size
 - o Algorithm used
 - o Time complexity
 - o Expected performance
- 2. Analyze why some algorithms become impractical as data size grows
- 3. Discuss how poor algorithm selection impacts:
 - o Response time
 - o Resource consumption
 - o User experience

Task 5: Recommendation & Decision Making

Based on your investigation:

- 1. Recommend the most suitable sorting and searching algorithm for:
 - o Daily order processing
 - o Real-time customer query handling
- 2. Justify your decision using complexity analysis, not intuition

Reflection Component (Mandatory)

Students must write a **reflection (200–300 words)** addressing:

- How algorithm complexity influences real-life system scalability
- Why average-case analysis is often more meaningful than worst-case
- What risks arise when inefficient algorithms are used in large systems
- How this investigation changed your understanding of algorithm design decisions

Assessment Rubric

Criteria	Marks
Problem understanding & data modeling	20%
Complexity analysis accuracy	30%
Comparative investigation & reasoning	25%
Real-life relevance & justification	15%
Reflection quality	10%

Guidelines:

Students must submit a **printed hard copy** of the assignment within the specified deadline. The submission should be **well-organized, properly formatted, and complete**. **Late submissions or soft-copy submissions will not be accepted.**

Any copied or plagiarized work will be rejected, and appropriate disciplinary action will be taken in accordance with academic regulations.

Submission Deadline: 01.02.2026